

Online Li-Ion Battery State-of-Health Estimation During Electric Vehicles Fast Charging: A Comparative Analysis of Advanced Data-Driven Techniques

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ABSTRACT Online battery state-of-health (SoH) assessment is paramount to improving battery electric vehicles (BEVs) competitiveness. By continuously acquiring data from the on-board battery, it enables to implement strategies aiming to reduce degradation rate and enhance BEVs safety. This work presents the application and validation of two novel patented methodologies for lithium-ion battery SoH estimation, evaluated under realistic BEVs fast-charging operating profiles and benchmarked against experimental measurements. The proposed methods, i.e., the “OCV–SoC-based method” relying on the open-circuit voltage (OCV)–state-of-charge (SoC) characteristic curve and the “discrete wavelet transform (DWT)-based method” applying DWT analysis to *in operando* voltage signals, are compared with the widely adopted reference technique incremental capacity analysis (ICA). Performances of all applied methods are assessed with reference to experimental capacity fading data obtained from a dedicated cycling aging campaign performed on commercial nickel manganese cobalt (NMC) lithium-ion cells. Both proposed methods exhibit a high accuracy (0.63% and 1.37% root-mean-square-error values for “DWT-based method” and “OCV–SoC-based method,” respectively) comparable with the ICA one (0.98%). Pros and cons of the proposed methods are discussed, together with future activities needed to extend their application domain.

INDEX TERMS Battery electric vehicles (BEVs), fast charging, lithium-ion battery (LIB) degradation, online assessment.

I. INTRODUCTION

The electrification of transportation sector aims to boost the clean energy transition [1]. Lithium-ion batteries (LIBs) represent the mainstream technology to power electric vehicles, thanks to fast response, high efficiency, high power and energy density, and low self-discharge [2], [3]. Electric vehicles integrating LIBs are generally classified as battery electric vehicles (BEVs). Nevertheless, the current challenges in developing low-cost and extended range batteries for electric vehicles hinder the diffusion of BEVs [4]. To create more compact and lightweight next-generation LIBs, research focuses on developing and optimizing cell active materials

to improve fundamental properties such as energy density, cycle life, and safety [5]. LIBs limited energy density still remains the major issue to BEV widespread (i.e., about 250–350 Wh/kg, significantly lower than theoretical limits). Furthermore, research efforts deal with manufacturing high-performance batteries by preventing the aging to be economically and ecologically viable [4]. For instance, the requirements of BEVs rapid charging times collide with a fast battery degradation, since LIBs are very sensitive to abusive operating conditions (e.g., overcharging, overdischarging, and elevated operating temperatures) [6]. Other factors, such as electrolyte breakdown, dendrite growth, and mechanical stress

in the electrode materials, result in capacity loss after 1000–3000 charge cycles and internal resistance increase, which limits LIBs long-term reliability [3], [7], [8]. Hence, BEV uncontrolled operating conditions can lead to unstable chain reactions that enhance degradation and could also cause a thermal runaway with the consequent fire and explosion of the battery. Therefore, understanding the electrochemical reactions and the material properties is crucial for battery safety assessment. Parameters such as voltage, temperature, and cathode material represent the key factors to control battery reactions [9].

To make BEVs competitive with internal combustion engine vehicles, it is necessary to continuously acquire data from the on-board battery to reduce degradation rate and enhance safety. Commonly, to ensure that BEV is operated within safe limits, on-board batteries are controlled via a battery management system (BMS), which provides information as measured voltage and current, pack and cell temperature, and estimated capacity of the battery [6]. Nevertheless, BMSs often use a low-cost microprocessor that hinders the use of best-in-class algorithms for the optimum operation of the batteries. Generally, to assess battery aging, BMS estimates battery state of health (SoH) through simple methods, such as coulombic counting [10], [11]. This method is not reliable and accurate for BEVs since, due to LIB limited depth-of-discharge (DoD) for safety reasons, the BMS computes SoH basing on partial charge/discharge cycles. Moreover, this method is strongly affected by measurement noise and faults of the data acquisition system. For BEV applications, LIBs end of life is fixed at the achievement of 80% of the initial battery capacity [12]. Therefore, without suitable monitor or prognostic solutions for evaluating capacity fading, BEV power and energy capabilities will degrade fast [13].

Accordingly, an accurate real-time SoH prediction is essential to guarantee LIB safety and reliability. In this context, to predict cycle and calendar aging, both model-based and data-driven methods have been implemented for SoH estimation [14]. For instance, model-based methodologies for LIB SoH estimation and remaining useful life (RUL) prediction have been investigated in literature [15], [16], [17]. On the other hand, noteworthy studies on data-driven methods are illustrated in the following. As example, the authors in [18] and [19] present data-driven approaches for battery performance monitoring and RUL estimation, respectively, while Carkhuff et al. [20] develop a BMS including real-time measurements based on multifrequency impedance to efficiently identify cell mismatches and emerging failures. In [21], a sequence-to-sequence model aims to predict the electrochemical impedance spectra after battery degradation. The strength of the model lies in the use of partial capacity sequences from a limited number of cycles, easily adaptable to different life stages in an end-to-end manner, but it is not applied for SoH online assessment. Gaussian process regression models are used to predict LIBs calendar and cyclic aging in [13] and [22]. An accurate and highly generic integrated framework of aging mechanisms and data-driven methods for LIB-accelerated aging diagnosis is demonstrated in [23]. In

[24], [25], [26], and [27], different data-driven machine learning models and neural networks are implemented to predict LIBs cycle life. However, many of these methods require high computational burden and a large dataset for an accurate and reliable prediction. Therefore, it is evident the complexity to implement heavy computational methods such as neural networks, Kalman filters, or fuzzy logic into the BMS for estimating battery SoH in BEVs [10].

This article aims to assess two ease-of-implementation data-driven methods, developed and patented by the authors, for battery SoH online monitoring into BEV application. Specifically, as innovative contribution to the literature, this article aims to validate the application of both methods on voltage profiles acquired during BEVs fast charging.

With reference to this application, the performance of the two proposed innovative methods is compared to SoH evolution deduced from experimental capacity measurements, as well as with SoH estimation provided by the incremental capacity analysis (ICA) method widely applied in the literature.

The ICA method relies on the analysis of the differential capacity curve obtained during constant-current (CC) charging or discharging, which reflects phase transitions and lithium intercalation processes occurring within the active electrode materials. The method is selected to provide a performance reference in estimating SoH, since it represents a well-established electrochemical diagnostic technique for LIBs, extensively employed for degradation analysis and SoH estimation over the past two decades. Seminal studies [28], [29], [30] demonstrated that variations in incremental capacity peak position, amplitude, and area can be directly correlated with capacity fade and specific degradation mechanisms, such as loss of lithium inventory and loss of active material. These works established ICA as a physically interpretable diagnostic tool capable of linking macroscopic performance degradation to electrode-level phenomena. Beyond degradation mechanism identification, ICA has also been widely adopted for quantitative SoH estimation. In [31] and [32], the concept of regional capacity is introduced, showing that SoH can be expressed as a linear function of the charge exchanged within a properly selected voltage interval around a dominant peak, enabling computationally efficient implementation suitable for BMSs. Similarly, in [32], the feasibility of implementing ICA-based SoH estimation in embedded systems for lithium iron phosphate (LFP) cells under electric vehicle operating constraints is demonstrated. More recently, Jenu et al. [33] validated ICA-based approaches for LFP, NMC, and lithium titanate oxide (LTO) chemistries under partial charging conditions and moderate current rates, reporting estimation errors typically below 2%–3% when appropriate voltage windows are selected.

ICA offers clear advantages, including strong physical interpretability, direct correlation with electrochemical processes, and compatibility with standard CC charging data without requiring complex battery models. Nevertheless, its derivative formulation makes it sensitive to measurement noise and operating conditions such as current rate, temperature, and DoD, which may affect robustness in dynamic

applications [33]. Despite these limitations, when implemented with appropriate voltage window selection and signal processing, ICA remains a validated and reliable benchmark for capacity-based SoH estimation. For this reason, it is adopted in this work as a literature reference method, providing a consistent baseline for comparison with the proposed approaches.

The first proposed innovative method for SoH online monitoring during BEVs fast-charging, named “OCV–SoC-based method,” consists in an innovative approach that relies solely on the knowledge of the battery specific open-circuit voltage (OCV)–state-of-charge (SoC) characteristic curve. The method, protected by the patent in [34], has not previously been disclosed in the scientific literature. It is based on two main assumptions, which are as follows:

the definition of SoC as a function of the actual available capacity, and therefore, explicitly dependent on SoH;

the invariance of the OCV–SoC characteristic curve with respect to aging when SoC is defined according to the previous point.

Although the latter assumption is inherently empirical and must be experimentally validated for each battery technology, it has been observed across different lithium-ion chemistries by the authors of the patented method. In this study, its validity is experimentally verified for the battery chemistry under investigation.

The second method (“discrete wavelet transform (DWT)-based method”), protected by the patent in [35], concerns the application of a specific algorithm providing SoH estimation in function of the number of charging/discharging cycles and peculiar features of the voltage signal obtained through its online analysis via DWT according to a suitable developed procedure. It is emphasized that the voltage profile processed via DWT is acquired during normal BEV operation, thus under a generic variable current pattern without the need for offline periods for LIB capacity determination, impedance, or OCV measurements. DWT is an ease-of-implementation and powerful tool facilitating the extraction of electrochemical features from signals with a broad frequency range by means of multiresolution analysis (MRA). Several papers applied DWT directly on voltage profiles [36], [37], [38], [39], [40]. In [41] and [42], some of the authors presented for the first time the application of DWT on typical BEVs test drive cycles to predict LIB RUL. Moreover, in [43], the same research group proved the validity of the DWT-based method application to predict RUL of sodium-ion chemistry. With reference to these studies, this work extends the application of the proposed method to the voltage profile acquired during a partial fast-charge process under typical current profiles, providing a useful innovative contribution to the literature.

Globally, this article significantly contributes to the literature by applying two innovative and ease-of-implementation methods (no expensive on-board equipment—e.g., for impedance measurements—and complex data elaboration algorithms are required) for online assessment of battery SoH

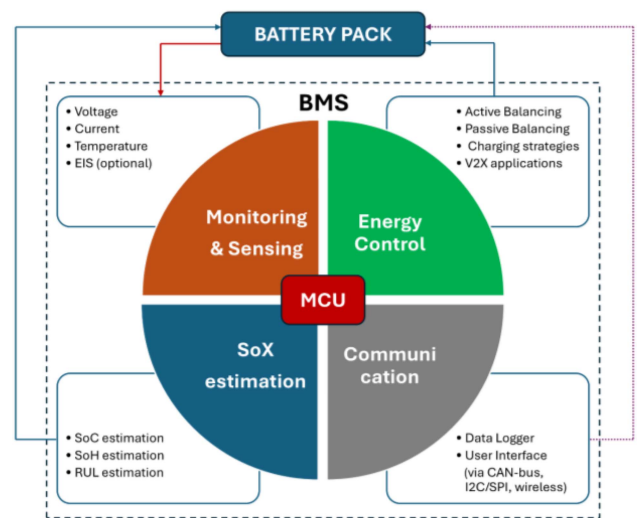


FIGURE 1. General schematic BMS flowchart.

during BEVs fast-charging. To perform such a comparative analysis, a suitable experimental campaign was performed on three LIB cells, each one aged over 1000 partial ch-/discharge cycles, measuring also the residual cell capacity at regular intervals. Specifically, each cycle is characterized by a charging current pattern typical of BEVs fast charging and CC discharge. At the same regular intervals, all the methods cited previously were applied to measure data gathered during the fast charge. The ICA method provides an SoH estimation accuracy [in terms of root mean square error (RMSE)] in line with the best performance presented in literature. Similar performances were obtained for the novel methods here proposed, validating their applicability also to BEVs fast-charging, while providing advantages in terms of tolerance to noise measurements, data acquisition faults affecting incremental current counting, and sensitivity to operating conditions such as current rate and DoD.

II. BMS INTEGRATION INTO ELECTRIC ON-BOARD SYSTEMS

The BMS plays a key role in managing batteries in safety operating conditions. As mentioned in the Introduction, the BMS continuously acquires data from the battery (e.g., voltage, current, and temperature) with sampling frequencies in the order of kilohertz and transfers data to the BEVs data logger and user interface via controller area network (CAN)-bus, wireless, or other communication protocols [44]. Other functionalities deal with the battery energy control (active and passive cells balancing, charging strategies, as well as innovative strategies for bidirectional Vehicle-to-X operation [45], [46]) and battery status estimation (i.e., SoC, SoH, and RUL). Fig. 1 illustrates a schematic flowchart of the BMS with details on hardware components, interface with the battery pack and its functionality. As visible in Fig. 1, the BMS generally consists of a microcontroller unit with limited non-volatile memory [47]. This hinders to store a great amount

TABLE 1. Specifications of the Lithium Cells Under Analysis

Typology	Lithium NMC	Photo
Geometry	Cylindrical 18650	
Nominal capacity (Ah)	2	
Operating voltage (V)	2.5 / 4.2	
Max discharge current (A) (continuous)	22	
Max charge current (A) (continuous)	4	
Operating temperature (°C) (discharge)	-20 / +75	
Operating temperature (°C) (charge)	0 / +50	

of data needed for estimating SoH through methods based on artificial intelligence or machine learning for an accurate SoH estimation. On the other hand, the methods proposed in this article, for real-time SoH assessment during BEV fast charging, require few data to be stored, making viable their implementation to on-board BMSs.

III. MATERIALS

The lithium-ion cells under test are 18 650 NMC cells having the specifications reported in Table 1. Details of the laboratory experimental setup are shown in Fig. 2. To control the temperature, the lithium cells under investigation were placed in a temperature-controlled environment at 20 °C, using thermocouples applied both in the test environment and on the external surface of the cells. The cells were then connected to a programmable battery cycler (NEWARE BTS4000-5V20A) to set the defined charge and discharge profiles within current (± 20 A) and voltage (0–5 V) ranges.

IV. DESIGN OF EXPERIMENTS AND TESTING PROCEDURES

Three 18650 NMC cells were aged through a specific experimental procedure, as outlined in Fig. 3. This procedure consists of two main phases, described as follows.

- 1) Phase 1, corresponding to the capacity determination of the tested cell. A full charge–discharge cycle is performed to determine the current capacity of the cells, aiming to measure cell capacity fading as function of number of cycles performed. Specifically, it consists of the following steps:
 - 2) a CC discharge at C/4 (0.5 A) until the minimum voltage (2.5 V) is reached, corresponding to SoC = 0%;
 - 3) full charge at C/2 (1 A) CC up to 4.2 V, followed by a constant voltage (CV) with 0.3 A as current cut-off (corresponding to SoC = 100%);
 - 4) full CC discharge at C/4 until the minimum voltage (i.e., 2.5 V) corresponding to SoC = 0%. The discharged capacity is then used to correlate to the current aging level;
 - 5) repetition of full charge to ensure the next protocol starts with the cell at SoC = 100%.

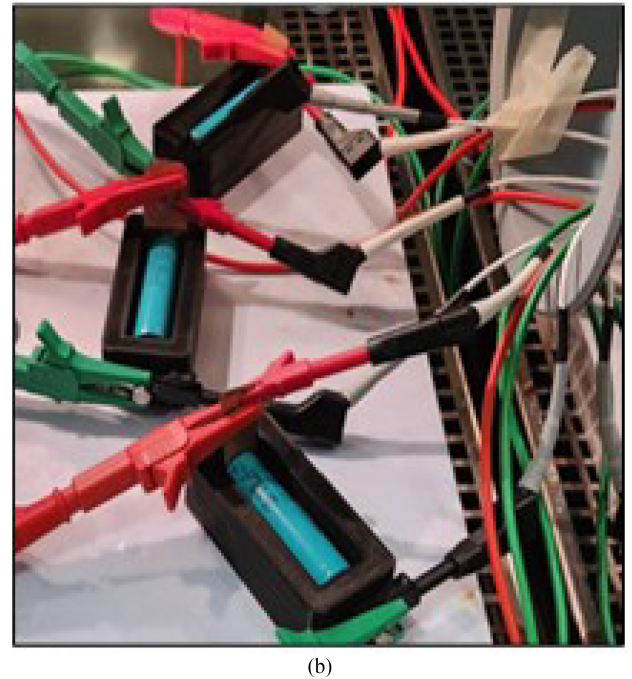
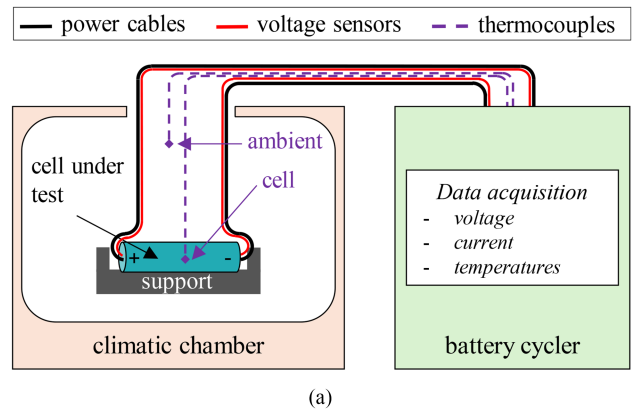


FIGURE 2. (a) Schematization of the laboratory setup: cell under test placed inside a climatic chamber and connected to battery cycler; acquisition of voltage, current, ambient, and cell temperature. (b) Photo of the cells under test inside the climatic chamber.

As final output of this protocol, the extracted capacity from the complete discharge of step c) is used to calculate the actual aging level, through the SoH indicator of the cells.

$$\text{SoH} = \frac{Q_{M-1}}{Q_0}. \quad (1)$$

This is defined, according to the aforementioned equation, by dividing the current extracted capacity (Q_{M-1} , where M is the number of “cycle aging” repetitions as indicated in Fig. 2) by the beginning-of-life (BoL) capacity value (Q_0), which corresponds to the capacity measured during the first performed protocol.

- 1) Phase 2 “Cycle aging”: This phase represents the actual stage in which the examined cells is aged by performing 100 charge/discharge cycles. To implement this phase,

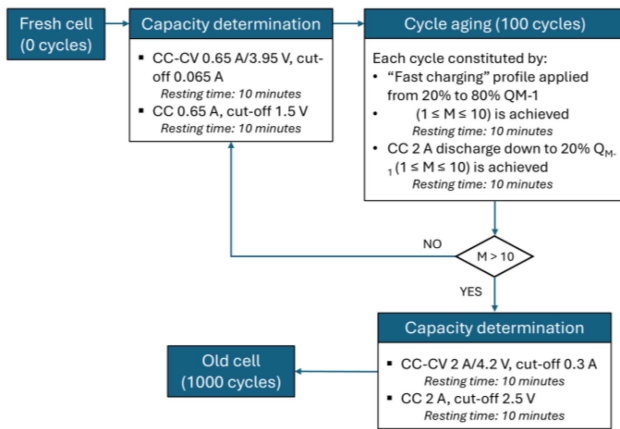


FIGURE 3. Flow chart diagram of the experimental cycle aging procedure.

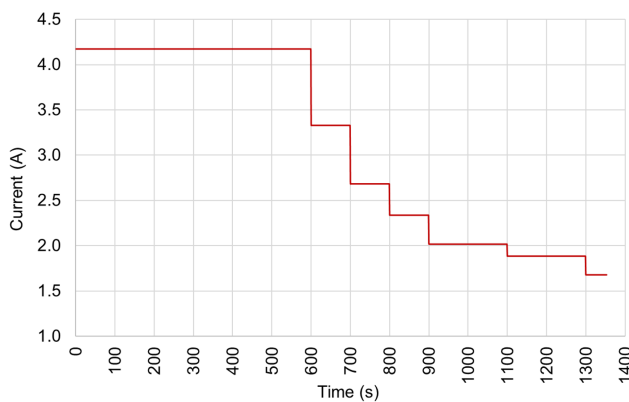


FIGURE 4. Implemented fast-charge profile during cycle aging.

it is necessary to define the operational specifications of the imposed charge/discharge cycles, including the current profile used for charging and discharging, the SoC range in which these profiles are executed, and the operating temperature (maintained at 20 ± 1 °C). Specifically, a fast-charging profile of a commercial electric vehicle [21], capable of recharging 60% of SoC in around 18 min, was taken into account to define the current profile used for charging, as shown in Fig. 4, while a constant discharging current of 2 A (1C) was applied. With regard to the operating SoC range, a 20%–80% window was considered. Therefore, at each repetition of the cycle aging, charged and discharged capacity are updated in consideration of the current full measured capacity (Q_{M-1}), as evident in Fig. 3. These conditions were chosen based on the typical usage of the examined cells in the context of battery applications for the automotive sector.

As final output, as outlined in Fig. 3, the completion of the experimental cycle ageing tests involves repeating ten times phases 1 and 2, ensuring 1000 total cycles and ten capacity measurements Q_{M-1} , $M = 1, \dots, 10$ (expressed in terms of SoH). Finally, a final capacity measurement is carried out to

characterize the aging level of the “old cells.” Throughout the test, cell voltage and current, as well as cell surface and ambient temperatures, were continuously measured.

V. METHODS FOR SOH ONLINE ASSESSMENT

A. OCV-SOC-BASED METHOD

The OCV–SoC-based method here illustrated is an innovative approach for estimating the aging level of a lithium battery relying exclusively on the knowledge of the battery-specific OCV–SoC characteristic. The authors state that this method is protected by a patent [34] and has not previously been disclosed in any scientific publication.

The OCV–SoC curve represents the well-known relationship between the battery OCV and its SoC. The proposed methodology is founded on two fundamental assumptions.

The first assumption concerns the definition of SoC. Unlike the conventional formulation, where SoC is computed referring to a fixed nominal capacity, the denominator is defined as the *actual available capacity* at the considered aging state. Accordingly, the SoC definition incorporates the battery SoH, as expressed in (2). In this formulation, SoC_0 denotes the initial SoC, $i(t)$ is the battery current (assumed positive during discharge), and Q_0 is the BoL capacity, consistently with the SoH definition provided in (1). As a consequence, the resulting SoC becomes an explicit function of SoH.

The second key assumption is that, under the adopted SoC definition, the OCV–SoC curve is invariant with respect to battery aging, i.e., it does not explicitly depend on SoH. The authors are fully aware that this represents a strong assumption. Its validity is inherently empirical and can only be assessed through experimental verification. While such invariance cannot be claimed a priori for all lithium-ion chemistries and operating conditions, it has been experimentally observed and validated by the authors of the patented method across different LIB technologies, as reported in [34]. Nevertheless, the general applicability of this hypothesis must be assessed on a case-by-case basis. For this reason, within the scope of this work, the assumed invariance of the OCV–SoC relationship with respect to aging is experimentally verified for the specific battery under investigation. The results shown in Fig. 5, derived from previous experimental activities conducted by the authors on cells identical to those employed in this study, provide supporting evidence.

Fig. 5 reports the OCV–SoC curves, where SoC is defined according to (2), for different aging levels ranging from $SoH = 100\%$ down to $SoH = 84\%$. The validity of the invariance assumption is supported by the substantial overlap of the curves across the considered SoH range. Furthermore, Fig. 5 also depicts, for each aging level, the deviation of the OCV values with respect to the beginning of life reference curve ($SoH = 100\%$). Such deviations are confined within a limited range of approximately 0–40 mV over the entire SoC domain shown. It can also be observed that, within specific SoC intervals (particularly in the ranges 70–100%

OCV-SoC curves and OCV differences with respect to BoL as function of SoH

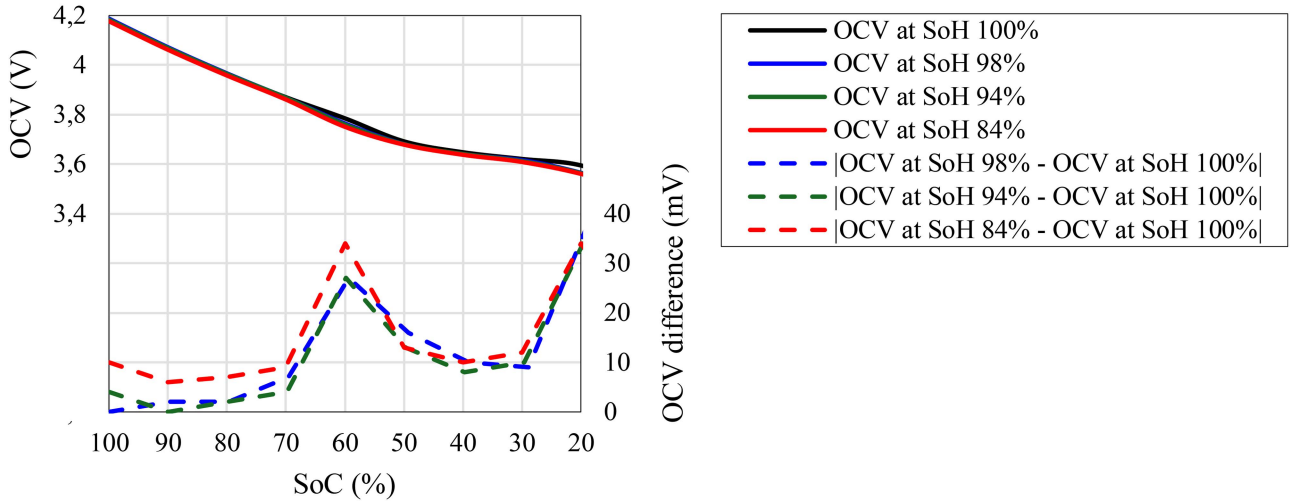


FIGURE 5. OCV-SoC curves and deviations with respect to the BoL reference as a function of SoH for a lithium-ion cell identical to the one used in this study.

and 30-50%) the voltage deviation remains even more tightly bounded.

These experimental findings support, for the considered battery technology and aging window, the practical validity of the invariance hypothesis, thereby ensuring methodological consistency and technical rigor. Under these conditions, the only information required to calibrate the proposed methodology is the OCV-SoC curve measured at beginning of life, i.e., for $\text{SoH} = 100\%$. Owing to the assumed and experimentally supported invariance with respect to SoH, the same curve can be reliably employed for lower SoH levels.

The analytical derivation of the SoH estimation is summarized as follows. By applying (2) over two distinct time instants, corresponding to two SoC values, (3) is obtained, where SoC_1 and SoC_2 represent the SoC at times t_1 and t_2 , respectively. Through straightforward algebraic manipulation, SoH can be explicitly derived as shown in (4) and (5). This leads to a compact expression for SoH that requires only the measurement of the current, the BoL capacity, and a pair of SoC values SoC_1 and SoC_2 . From an operational standpoint, it is essential that these SoC values be directly obtained from the OCV-SoC curve, meaning that they correspond to two measured OCV values under equilibrium conditions. Accordingly, the final SoH formulation is expressed in (6), where the SoC values are replaced by the function $\xi(\cdot)$, representing the one-to-one OCV-SoC relationship.

From a more intuitive standpoint, the physical meaning of the method can be summarized as follows. Owing to the adopted SoC definition, where SoC is explicitly formulated as a function of SoH through the actual available capacity, the OCV-SoC relationship becomes independent of aging. Under this condition, a given variation in OCV corresponds, through the invariant OCV-SoC mapping, to a fixed SoC interval regardless of the degradation level. Consequently, the amount

of charge exchanged between two equilibrium OCV states over that fixed SoC interval is directly proportional to the actual available capacity of the battery. As aging progresses and the available capacity decreases, the same OCV (and thus, SoC) interval corresponds to a progressively smaller quantity of exchanged charge. Therefore, by measuring the charge required to move between two OCV levels and relating it to the BoL reference capacity, the reduction in available capacity, and hence, the SoH, can be directly inferred. In essence, the method estimates battery aging by exploiting the fact that, for identical OCV intervals, the battery stores or delivers a decreasing amount of charge as degradation advances.

$$\text{SoC} = \text{SoC}_0 - \frac{\int i(t) dt}{\text{SoH} \cdot Q_0} \quad (2)$$

$$\text{SoC}_2 = \text{SoC}_1 - \frac{\int_{t_1}^{t_2} i(t) dt}{\text{SoH} \cdot Q_0} \quad (3)$$

$$\text{SoC}_2 = \text{SoC}_1 - \frac{\int_{t_1}^{t_2} i(t) dt}{\text{SoH} \cdot Q_0} \quad (4)$$

$$\text{SoH} = \frac{\int_{t_1}^{t_2} i(t) dt}{Q_0 \cdot (\text{SoC}_1 - \text{SoC}_2)} \quad (5)$$

$$\text{SoH} = \frac{\int_{t_1}^{t_2} i(t) dt}{Q_0 \cdot (\xi(\text{OCV}_1) - \xi(\text{OCV}_2))} \quad (6)$$

B. DWT-BASED METHOD

The registered voltage of an electrochemical storage device is a nonstationary signal. Hence, the Fourier analysis represents a not useful method due to the loss of time information into the frequency domain passage [48]. In this context, wavelet transform can capture both frequency and location in time information, by splitting a signal into shifted and scaled versions

of the original wavelet [49]. Generally, the wavelet transform is distinguished into continuous (CWT) and discrete (DWT) wavelet transform. DWT analysis is based on wavelets discretely sampled, decomposing the original signal in different resolutions, to reduce CWT complexity [50] due to the high amount of data to be processed.

Thus, the instantaneous variations of the signal can be analyzed in all the frequency bands, making the DWT technique to be very effective in systems diagnosis [49]. Basically, the original signal is divided into different time–frequency scales by passing through a series of high-pass filters with different cut-off frequencies to examine the high frequencies, while the signal low frequencies are obtained applying a series of low-pass filters at the original signal. This approach constitutes the MRA, enabling the separation of the low frequency (approximated structure) from the high-frequency content (detailed structure) of the studied signal [51].

Considering n decomposition levels, the wavelet function $\xi(t)$ and the scaling function $\vartheta(t)$ are expressed by

$$\xi_{j,n}(t) = 2^{\frac{j}{2}} \sum_n w_{j,n} \xi[2^j t - n] \quad (7)$$

$$\vartheta_{j,n}(t) = 2^{\frac{j}{2}} \sum_n u_{j,n} \vartheta[2^j t - n] \quad (8)$$

where w_j and u_j represent the scaling coefficient at scale j and the wavelet at scale j , respectively. The MRA mathematical representation is expressed as follows [52]:

$$V_j = W_{j+1} \oplus V_{j+1} = W_{j+1} \oplus \dots \oplus W_{j+n} \oplus V_n \quad (9)$$

where

- 1) V_{j+1} is the approximate version of the input signal at scale $j+1$;
- 2) W_{j+1} is the detailed version, including all the transient phenomena of the input signal at scale $j+1$;
- 3) n represents the decomposition level;
- 4) $\oplus$ stands for the sum of the two decomposed signals.

Assuming a constant time intervals sampling, the original signal $x_j(t)$ at scale j is $x_j(t) = (v_0, v_1, \dots, v_{N-1})$. The sampling number is equal to $N = 2^j$, where j is an integer number. DWT computed for $x_j(t)$ is shown as follows, taken into account (11)–(13):

$$\begin{aligned} \text{DWT}(x_j(t)) &= \sum_k x_j(t) \xi_{j,k}(t) = \\ & 2^{\frac{(j+1)}{2}} \left(\sum_n uA_{j+1,n} \xi[2^{j+1}t - n] + \sum_n wD_{j+1,n} \vartheta[2^{j+1}t - n] \right) \\ & 0 \leq n \leq \frac{N}{2^j} - 1 \end{aligned} \quad (10)$$

with

$$uA_{j+1,n} = \sum_k c_{j,k} v_{j,k+2n} \quad 0 \leq k \leq \frac{N}{2^j} - 1 \quad (11)$$

$$wD_{j+1,n} = \sum_k w_{j,k} v_{j,k+2n} \quad 0 \leq k \leq \frac{N}{2^j} - 1 \quad (12)$$

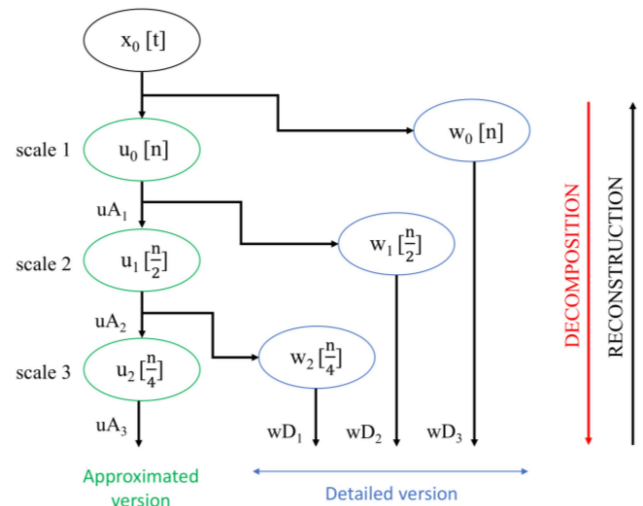


FIGURE 6. DWT basic decomposition and reconstruction view.

$$d_k = (-1)^k c_{2p-1-k} \quad p = \frac{N}{2^j} \quad (13)$$

where $uA_{j+1,n}$ and $wD_{j+1,n}$ represent the approximate and the detailed versions at scale $j+1$, respectively.

DWT basic decomposition and reconstruction is outlined in Fig. 6. Further details on the DWT procedure can be found in [41].

Finally, to calculate the energy distribution of the original signal, it is involved the application of the Parseval's theorem [52]. For instance, if a discrete signal $x(n)$ is the current flowing through a resistive element, the square sum of the spectrum coefficients (a_k) of the Fourier transform in the frequency domain represents the energy of the resistance, as indicated by

$$\frac{1}{N} \sum_{n=1}^N |x(n)|^2 = \sum_{n=1}^N |a_k|^2 \quad (14)$$

where N is the sampling period. Therefore, the energy distribution of the signal in DWT computed by Parseval's theorem comes from

$$\begin{aligned} \frac{1}{N} \sum_t |x(t)|^2 &= E_a + E_d = \\ &= \frac{1}{N_j} \sum_t |uA_{j,k}|^2 + \sum_{j=1}^J \left(\frac{1}{N_j} \sum_k |wD_{j,k}|^2 \right) \end{aligned} \quad (15)$$

where E_a and E_d are the average powers of the approximated form and the detailed form of the decomposed signal, respectively.

From previous works of the authors [41], [42], a method was developed and validated to assess in real-time LIB capacity fading. Specifically, (16) expresses the dependence of capacity fading (C_{fade}) of LIB on detail and approximation

coefficients (as resulting from the decomposition process depicted in Fig. 6), and number of operating cycles.

$$C_{\text{fade}}(A, D_{1,4}, N) = a \frac{A_0}{A_i} e^{-\left(b \frac{N_i^c}{A_i} \frac{\sum_{n=1}^4 \Delta_i}{\sum_{n=1}^4 D_{n,0}}\right)} \quad (16)$$

where

- 1) A_i and A_0 are the approximation coefficients related to i th aging cycle and to the fresh battery, respectively.
- 2) N_i is the corresponding i th number of cycles performed by the battery.
- 3) $\delta_i = \sum_i \sum_n D_{n,i}$ is the cumulative sum of the details' coefficients after i cycles of aging ($n=1 \rightarrow 4$).
- 4) $D_{n,0}$ represents the detail coefficients at n levels for the new battery.
- 5) a , b , and c are constant parameters to be assessed through a fitting procedure.

C. ICA METHOD

The ICA-based SoH estimation implemented in this work follows the regional capacity framework proposed in [31]. This formulation was selected as a literature validated and computationally efficient implementation of ICA, capable of directly linking measurable charging data to progressive capacity degradation without relying on complex electrochemical or equivalent circuit models, suitable for use in online applications, i.e., while the BMS is operating.

The ICA-based procedure is illustrated in Fig. 7. Starting from CC charging data, the incremental capacity curve is computed as

$$IC(V) = \frac{dQ}{dV} \quad (17)$$

where Q represents the accumulated charge obtained via Coulomb counting by

$$Q_k = \sum_{i=1}^k I_i \Delta t. \quad (18)$$

To mitigate numerical instability associated with direct differentiation, a finite-difference approximation is adopted as follows:

$$IC_k \approx \frac{Q_k - Q_{k-N}}{V_k - V_{k-N}} \quad (19)$$

where N is a smoothing window selected to balance resolution and robustness. Due to the derivative nature of ICA, the computed incremental capacity curve is filtered using a low-pass filter prior to feature extraction, to reduce amplification of measurement noise.

Fig. 7(a) depicts the filtered incremental capacity curve together with the corresponding charging curve as a function of voltage. This combined representation allows the dominant ICA peak and the associated charge variation to be visualized within the same voltage domain. Following [31], the dominant peak of the filtered IC curve is identified and its corresponding voltage V_{peak} is determined. A symmetric voltage interval

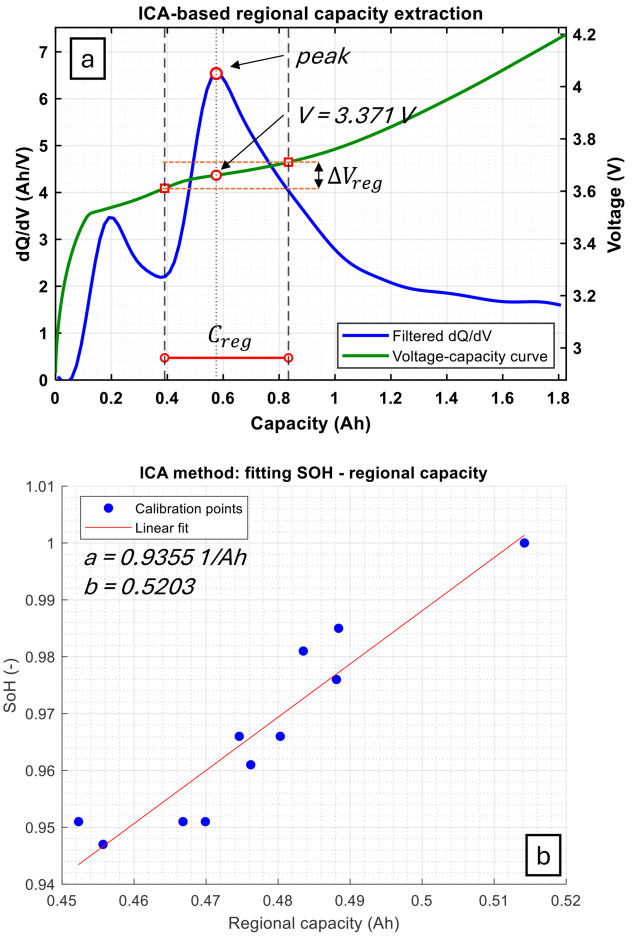


FIGURE 7. ICA-based SOH estimation procedure. (a) Example at 500 cycles cell #1 of filtered incremental capacity curve and voltage-capacity curve with the dominant ICA peak, selected interval $[V_0, V_1]$, and corresponding regional capacity C_{reg} . (b) Linear regression between regional capacity and SOH using the calibration points.

ΔV_{reg} centered around this peak is then defined as expressed as follows:

$$V_0 = V_{\text{peak}} - \frac{\Delta V_{\text{reg}}}{2} \quad (20)$$

$$V_1 = V_{\text{peak}} + \frac{\Delta V_{\text{reg}}}{2}. \quad (21)$$

The voltage limits V_0 and V_1 are then mapped onto the charging curve, as illustrated in Fig. 7(a), and the corresponding regional capacity is computed according to

$$C_{\text{reg}} = Q(V_1) - Q(V_0) \quad (22)$$

where $Q(V)$ is obtained from the charging curve via interpolation. This formulation captures the charge exchanged within a characteristic electrochemical region associated with the dominant phase transition, ensuring strong correlation with capacity fade while limiting sensitivity to local peak deformation.

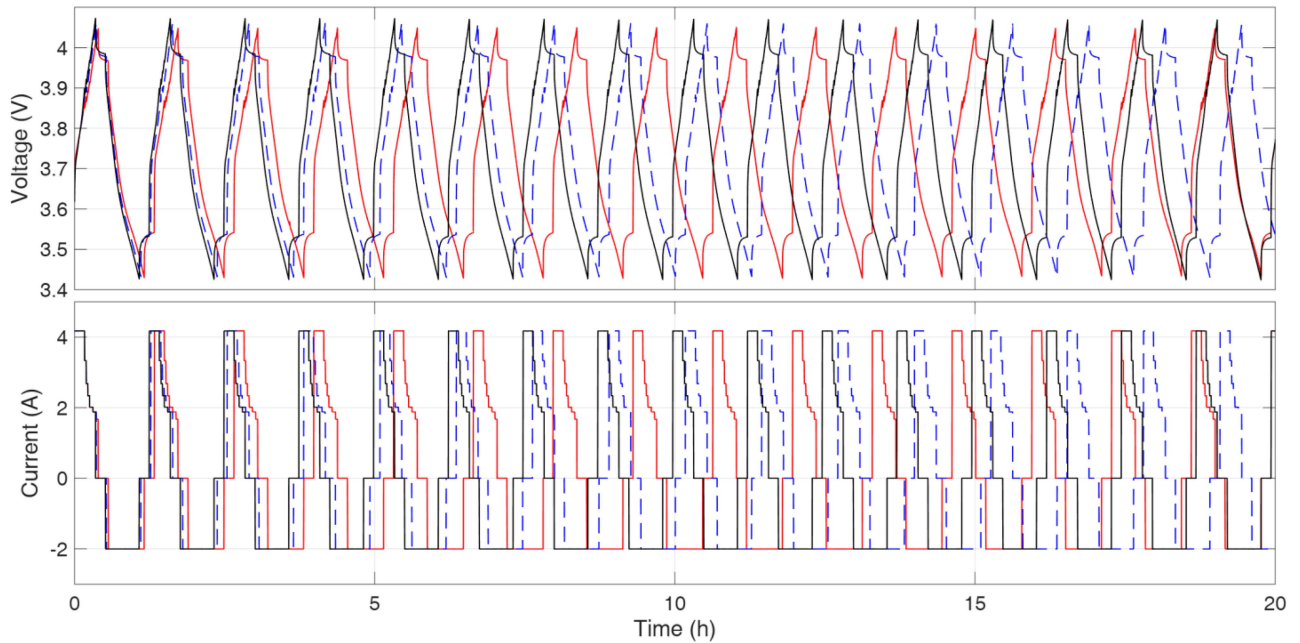


FIGURE 8. Measured voltage (upper figure) and applied current (lower figure) on cell 1 for cycles moving from 50 to 65 (red solid line), from 550 to 565 (blue dotted line) and from 950 to 965 (black solid line).

Consistent with [31], a linear relationship between regional capacity and SoH is assumed by

$$\text{SoH} = a \cdot C_{\text{reg}} + b \quad (23)$$

where the coefficients a and b are identified through least-squares regression. The resulting calibration curve is reported in Fig. 7(b), where the calibration points (performed using the experimental dataset obtained for the same cells type) and the corresponding linear fit are shown. The voltage window ΔV_{reg} is selected to maximize linear correlation while remaining compatible with the voltage range consistently available across all charging profiles.

The model calibration is performed using the experimental dataset previously obtained under controlled cycling conditions, ensuring that the ICA-based estimator is trained and validated on the same cells and operating scenarios considered for the evaluation of the presented methods.

VI. RESULTS AND DISCUSSION

A. EXPERIMENTAL RESULTS

Fig. 8 shows voltage and current evolution along test duration, with a focus on test fragments corresponding 50–65, 550–565, and 950–965 cycles.

It can be observed as the operating voltage range slightly increases due to degradation, while measured capacity value, averaged over the three cells, decreases from 1998.4 mAh at the beginning of the test to 1834.3 mAh. Specifically, minimum and maximum voltage values, detected over voltage evolution acquired at cell 1 terminals, move from 3.437–4.05 V at the 50th cycle to 3.432–4.061 V relative to 550th cycle, and to 3.42–4.065 V gathered at the 950th cycle. Fig. 9 depicts

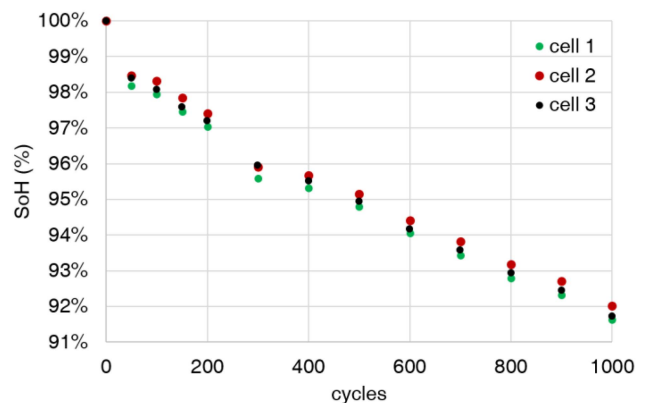


FIGURE 9. Measured capacity fading of tested 18 650 NMC cells.

the capacity fading trend measured for each cell. As evident from Fig. 9, SoH evolution is very similar among the three tested cells, showing a maximum deviation of 0.23% with respect to the SoH average values assessed at different cycle numbers. Nevertheless, data measured on all the tested cells are considered for the application of SoH estimation methods in the following, to prove their robustness and generalization capability.

B. APPLICATION OF SOH ONLINE ASSESSMENT METHODS TO EXPERIMENTAL DATASET

Figs. 10–12 refer to cell 1, cell 2, and cell 3, respectively. Each figure illustrates SoH data assessed based on measured capacity values, obtained from periodic reference capacity

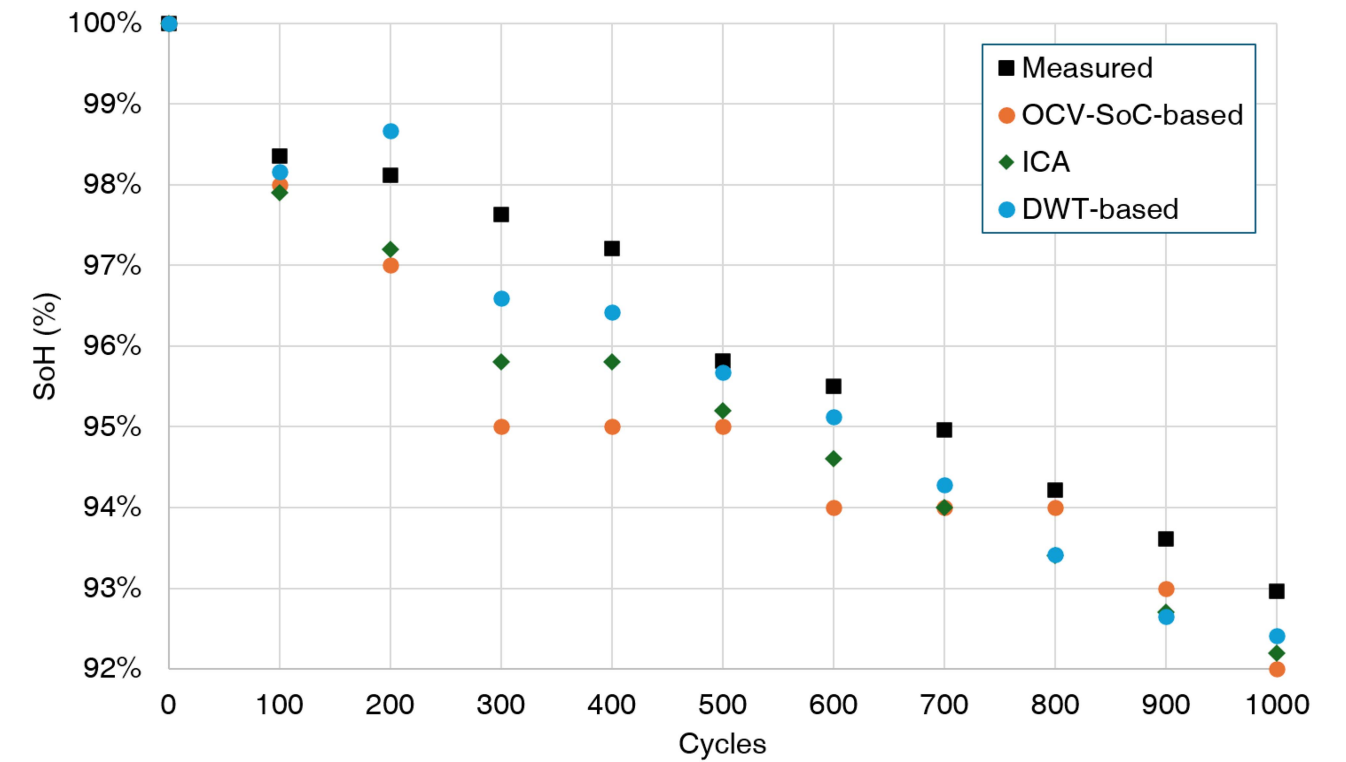


FIGURE 10. Evolution of the measured SoH (black) and predicted SoH over cycle aging by applying. (a) OCV-SoC-based method (orange), (b) DWT-based method (blue), and (c) ICA reference method (green) on cell 1.

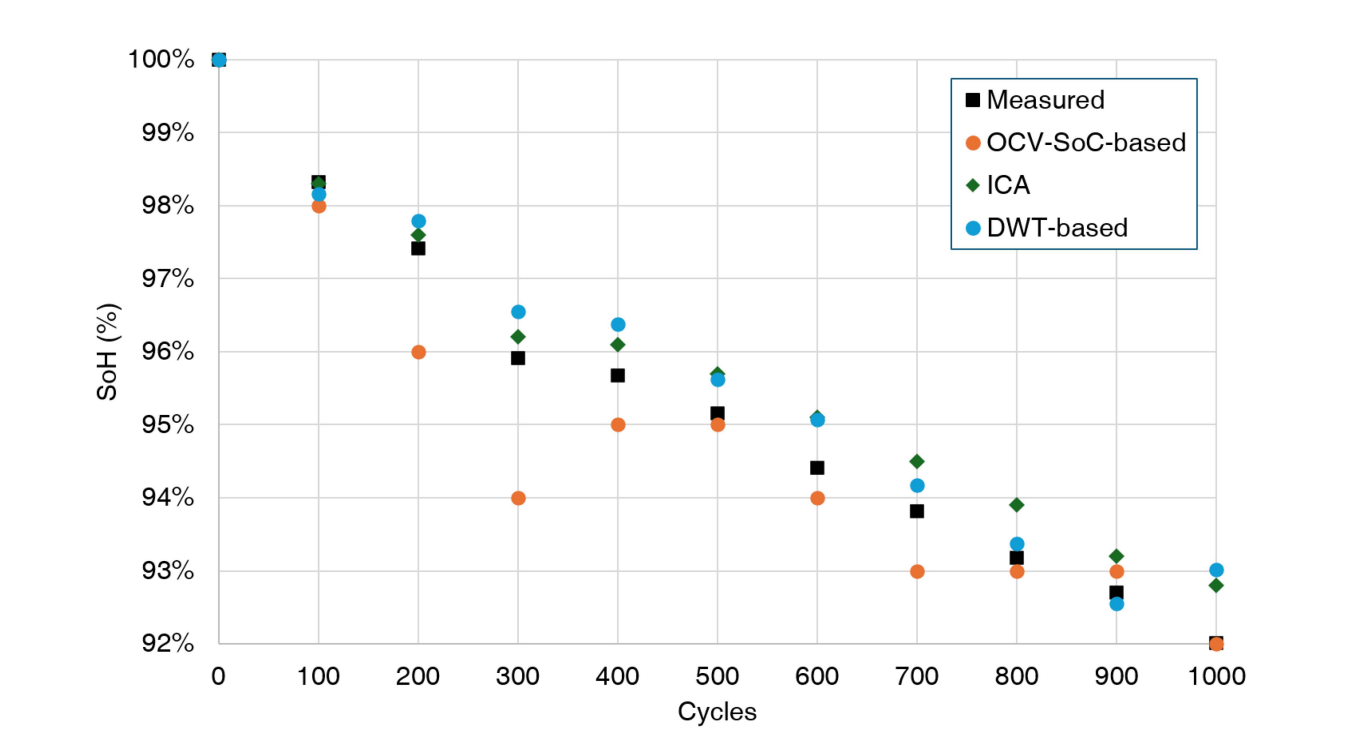


FIGURE 11. Evolution of the measured SoH (black) and predicted SoH over cycle aging by applying. (a) OCV-SoC-based method (orange), (b) DWT-based method (blue), and (c) ICA reference method (green) on cell 2.

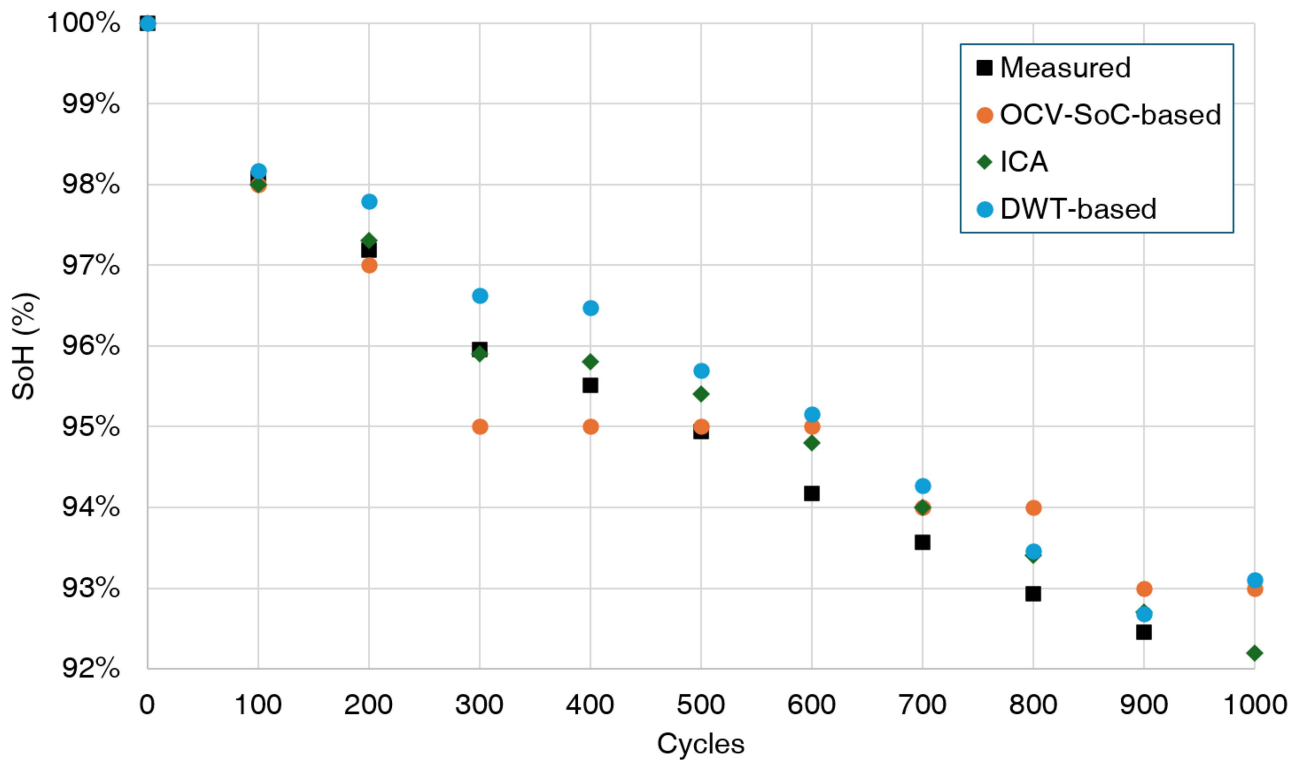


FIGURE 12. Evolution of the measured SoH (black) and predicted SoH over cycle aging by applying. (a) OCV-SoC-based method (orange), (b) DWT-based method (blue), and (c) ICA reference method (green) on cell 3.

TABLE 2. RMSE Values Obtained for Cell 1, Cell 2, and Cell 3 by Applying the Different SoH Estimation Methods

	OCV-SoC	DWT	ICA
cell 1	1.29%	0.73%	1.01%
cell 2	0.86%	0.51%	0.52%
cell 3	0.75%	0.61%	0.42%
mean	0.97%	0.62%	0.65%

tests on the specific cell (in black), together with the predicted SoH evaluated values obtained by applying the OCV-SoC-based method (orange), the DWT-based method (blue), and the ICA reference method (green). The discussion concerning the application of such methods and the obtained results, as summarized in Table 2, is presented as follows.

1) OCV-SOC-BASED METHOD

Estimated SoH values are computed according to the method described in Section V-A, exploiting pairs of OCV measurements and the invariant BoL OCV-SoC curve. A good agreement between measured and estimated SoH can be observed over the entire aging range investigated for all the tested cells. The proposed method accurately tracks the progressive capacity fading, correctly reproducing the degradation trend. The deviations between experimental and estimated values remain limited throughout the considered SoH window. According to Table 2, the resulting RMSE between measured and estimated

SoH is equal to 1.29%, 0.86%, and 0.75% for cells 1, 2, and 3, respectively, with a mean RMSE value slightly lower than 1%, confirming the good accuracy of the OCV-SoC-based method.

A closer inspection of Figs. 10–12 (see orange points) reveals that the third and fourth data points exhibit the most noticeable deviations from the overall trend, especially for cells 1 and 2. These represent localized discrepancies where the estimation departs more evidently from the experimental values; however, the maximum error remains within 3% and 2% for cells 1 and 2, respectively. This behavior can be interpreted as a minor limitation of the proposed approach, potentially related to localized variations not fully captured by the invariance assumption. Nevertheless, such deviations remain confined and do not significantly affect the global accuracy or the capability of the method to reproduce the overall degradation trajectory.

Overall, these results demonstrate that, under the experimentally validated invariance assumption of the OCV-SoC relationship, the proposed approach provides a reliable estimation of battery aging while requiring only OCV measurements at the beginning of life and current integration.

2) DWT-BASED METHOD

Predicted values are computed according to (16), considering a , b , and c parameters equal to 0.96, 0.06 and 0.43, respectively. The resulting RMSE corresponds to 0.73%, 0.51%,

and 0.61%, for cell 1, cell 2, and cell 3, respectively, with an average value equal to 0.62%. Regarding the tuning of the parameters, it is highlighted that a and b coefficients values resulted to be the same obtained by the authors in [41] and [42] for LIBs, although on a different Li-ion cathode chemistry (i.e., Nickel–Cobalt–Aluminum). With reference to the c parameter in (16), which has a direct effect on the cycles number impact, it is emphasized that, the reduced SoC window considered for cycle aging in this work (i.e., 60%, from 20% to 80% SoC) resulted in a reduced parameter value (0.43 versus 0.7) with respect to the previous outcomes presented in [41] and [42]. As matter of fact, the latter considered 100% as SoC window. Hence, in this study, at parity of aging cycles, this results in a lower capacity fading, requiring a lower c coefficient according to (16).

3) ICA METHOD

SoH values are estimated for cells 1, 2, and 3 through the ICA-based regional capacity method, used as reference SoH estimation method (green points in Figs. 10–12). Obtained results correspond to RMSE between measured and estimated SoH equal to 1.01%, 0.52%, and 0.42% for cells 1, 2, and 3, respectively, confirming the consistency of the ICA-based reference method on the investigated dataset. Mean RMSE equals 0.65%, therefore, with a good accuracy with respect to measured SoH values.

Specifically, a clear agreement between measured and ICA-estimated SoH can be observed over the entire ageing range considered. The predicted values closely follow the experimental degradation trend, correctly reproducing both the approximately linear fading behavior and the gradual reduction in capacity with increasing cycle number.

VII. CONCLUSION

This work presents the application and validation of two novel patented methodologies for LIB SoH estimation, evaluated under realistic operating profiles and benchmarked against experimental measurements. The proposed approaches (referred to as the OCV–SoC-based and DWT-based methods) were compared with the widely adopted reference technique ICA, which is commonly considered a benchmark in the literature for battery aging diagnostics. The results discussed in this study are grounded on experimental data obtained from a dedicated cycling aging campaign performed on commercial NMC lithium-ion cells, ensuring that the assessment reflects realistic degradation behavior.

The ICA method achieves an SoH estimation accuracy with an average RMSE of approximately 0.65% over the three tested cells, with a maximum value of about 1%, which is consistent with the best performance levels reported in the state of the art. The two proposed methodologies demonstrate comparable accuracy, thereby confirming their validity as alternative SoH estimation techniques. Moreover, the proposed methods exhibit improved tolerance to measurement noise and faults of the data acquisition system.

Among the investigated approaches, the OCV–SoC-based method shows the capability to accurately reproduce the experimentally observed aging trend, achieving an average RMSE of 0.97% over tested cells. This level of accuracy can be regarded as a very promising outcome, as it validates the theoretical assumptions underlying the method and confirms its effectiveness in capturing the degradation evolution of the cell. An important advantage of this approach is its low calibration requirement: the method relies solely on the OCV–SoC characterization measured at BoL, it exhibits a reduced sensitivity to operating conditions, such as current rate and DoD, and does not require recalibration or parameter updates throughout the battery lifetime. This feature represents a significant benefit for real-world BMS deployment, as it reduces implementation complexity and eliminates the need for periodic reidentification procedures as the cell ages.

Results obtained by applying the DWT-based method show a high accuracy in SoH estimation, with an average RMSE of 0.62%. Moreover, as a point of strength, it is remarked that SoH assessment is performed under *in-operando* BEV conditions, without the need for offline periods, based on routine voltage measurements. It requires only few parameters to be stored, constituting an ease-of-implementation tool. With respect to the previous studies of the authors, such outcomes validate the method application to voltage evolution during fast-charging, exhibiting a reduced sensitivity to the specific load pattern characteristic of BEV operation. On the other hand, the method, previously validated on full ch-/discharge cycles, exhibits a high-accuracy when applied to partial cycles over a 60% SoC window (from 20% to 80% of the battery capacity), after a BoL calibration of the c parameter in (16). Aiming to further improve the method, a dedicated activity will be performed in the future to assess the variation of such a parameter, which modulates the impact of cycles number on capacity fading, varying the operating SoC window in a wider range. Under operating conditions characterized by a significant variation in the SoC window during fast charging over the battery's lifetime, parameter c should be tuned based on the mapping obtained at the battery's beginning of life.

The features and performance detailed previously make the proposed methods particularly attractive for practical implementation in BMSs, especially in applications involving fast-charging operation in BEVs.

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